Supplementary Materials for "Bayesian Inference for Incomplete 2×2 Diagnostic Tables"

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Abstract

This supplement contains five parts. First, section S1 applies the single-row reconstruction model of the main manuscript to the oral brush-biopsy study of Svirsky et al. [1] (Case 1, partial verification). Section S2 applies the joint model with known N to the intracranial hemorrhage study of Wismüller and Stockmaster [2] (Case 2, $N = 620$), first using the reported counts only and then adding the sensitivity, specificity and accuracy reported in that study. Section S3 gives the design and detailed results of the simulation study. Section S4 describes the exact computation used for all results, including the extension to unknown N, and the WinBUGS implementation, with WinBUGS results. Finally, section S5 gives full posterior summaries, including the hyperparameters, for the breast magnetic resonance imaging (MRI) application of the main manuscript, and the results with priors chosen by rule. Section numbers below without the prefix S refer to the main manuscript.

S1. Case 1: Partial Verification

Data

Svirsky et al. [1] compared computer-assisted oral brush-biopsy results with follow-up scalpel biopsy and histology. Among 243 patients with abnormal brush-biopsy findings and reported scalpel-biopsy and histology results, 93 were confirmed as having dysplasia or carcinoma (TP) and 150 were histology negative (FP). Verification results are reported only for the test-positive subgroup and the published report does not give counts for the test-negative row, which we therefore treat as unobserved (Table S1).

Table S1: Available and missing information from Svirsky et al. [1].

	Histology Positive	Histology Negative
Brush biopsy abnormal (test positive)	93 (TP)	150 (FP)
Brush biopsy normal (test negative)	not reported (FN)	not reported (TN)

The observed counts are TP = 93 and FP = 150. The denominators $n_1 = \text{TP} + \text{FN}$ and $n_2 = \text{FP} + \text{TN}$ are unknown, and there is no information on the total study size N. The

PPV can be computed directly as $93/243 \approx 0.38$, but the sensitivity and specificity are not identified from the published data alone.

Model

We apply the independent binomial-n reconstruction model of Section 4.1 of the main manuscript separately to each column of the observed row. For the diseased column ($y = \mathrm{TP} = 93$, $n = n_1$, p the sensitivity) and for the non-diseased column ($y = \mathrm{FP} = 150$, $n = n_2$, p the false positive rate), the model is

$$y \mid n, p \sim \mathrm{Bin}(n, p), \qquad n \geq y, \tag{S1}$$

$$p \sim \mathrm{Beta}(\alpha, \beta), \tag{S2}$$

$$n \mid p^\star, r \sim \mathrm{NegBin}(p^\star, r)\, \mathbf{1}\{n \geq y\}, \tag{S3}$$

$$r \mid \lambda \sim \mathrm{Poisson}(\lambda)\, \mathbf{1}\{r \geq 1\}, \qquad \lambda \sim \mathrm{Gamma}(a, b), \tag{S4}$$

$$p^\star \sim \mathrm{Beta}(\alpha^\star, \beta^\star). \tag{S5}$$

As in the main manuscript, $\mathrm{NegBin}(p^\star, r)$ is the number of failures before r successes with success probability $p^\star$, the gamma distribution uses the rate parameterization, and both truncations are normalized. The hyperparameters below are the illustrative values used for the breast MRI example in Section 6.1 of the main manuscript, not the default denominator priors of Step 3 in Section 4.4.

Diseased Stratum ($y = \mathrm{TP} = 93$)

Hyperparameters.

$$a = 1,\ b = 0.1, \qquad \alpha = 2,\ \beta = 1, \qquad \alpha^\star = 1,\ \beta^\star = 1.$$

The $\mathrm{Beta}(2, 1)$ prior on sensitivity has mean $2/3$ and favors at least moderate sensitivity while still allowing substantially smaller values. The $\mathrm{Gamma}(1, 0.1)$ prior on λ has mean 10 and variance 100, and the uniform $\mathrm{Beta}(1, 1)$ prior on $p^\star$ allows the negative-binomial prior on n_1 to adapt to the data. The posterior was computed exactly (Section S4) and we report the posterior summary in table S2.

Table S2: Posterior summary: Svirsky diseased stratum, single-row model (exact computation).

Parameter	Mean	SD	2.5%	Median	97.5%
λ	10.38	9.53	0.76	7.51	35.96
n_1	96.83	18.97	93.0	93.0	121.0
p	0.962	0.072	0.760	0.984	0.999
$p^\star$	0.546	0.269	0.075	0.552	0.978
r	10.41	9.92	1.0	7.0	37.0

The posterior mean of n_1 is approximately 97, with 95% credible interval $(93, 121)$. The posterior median is 93, the smallest value compatible with the observed count, which reflects the large posterior mass on high sensitivity. With a single binomial observation the data alone do not determine the sensitivity, and this concentration comes from the priors on p and n_1 together with the beta-binomial marginal likelihood, which decreases in n_1.

Non-Diseased Stratum ($y = \text{FP} = 150$)

Hyperparameters.

$$a = 2,\ b = 1, \qquad \alpha = 2,\ \beta = 5, \qquad \alpha^\star = 1,\ \beta^\star = 50.$$

The Beta$(2, 5)$ prior on the false positive rate has mean $2/7 \approx 0.29$, reflecting an expectation of moderate specificity. The Beta$(1, 50)$ prior on $p^\star$ has mean approximately 0.02 and places substantial mass on larger values of n_2, consistent with the common scenario in which the non-diseased population is larger than the diseased population. The posterior was computed exactly (Section S4). We present the posterior summary in table S3.

Table S3: Posterior summary: Svirsky non-diseased stratum, single-row model (exact computation).

Parameter	Mean	SD	2.5%	Median	97.5%
λ	2.483	1.504	0.481	2.178	6.195
n_2	514.3	453.4	201.0	395.0	1540.0
p	0.388	0.171	0.097	0.377	0.738
$p^\star$	0.011	0.009	0.001	0.008	0.036
r	2.959	2.018	1.0	2.0	8.0

The posterior for n_2 is wide, with a mean near 514, a median of 395 and a heavy right tail. This reflects the substantial uncertainty inherent in estimating an unknown denominator from a single count of 150 false positives with no bound on N. Without any external constraint on the total study size, the non-diseased denominator is very weakly identified by a single observation.

Reconstructed Table

Table S4 combines the posterior summaries of the two independently fitted strata, and Table S5 gives the posterior summaries of the derived measures. Because the two strata were fitted separately , the derived measures were computed from the product of the two posteriors for n_1 and n_2. Posterior means of the derived measures differ from ratios of posterior means of the cells, markedly so for the right-skewed non-diseased stratum, so the derived measures are summarized directly rather than computed from the cell means.

Table S4: Reconstructed 2×2 table for Svirsky et al. [1]. Entries in the negative row are posterior means with 95% credible intervals, not recovered counts. The overall total is not constrained.

	Histology Positive	Histology Negative
Brush biopsy abnormal	93 (TP, observed)	150 (FP, observed)
Brush biopsy normal	3.8 (0, 28) (FN)	364.3 (51, 1390) (TN)
Total	n_1: 96.8 (93, 121)	n_2: 514.3 (201, 1540)

Table S5: Posterior summaries of the derived measures for Svirsky et al. [1], computed from the product of the posteriors for n_1 and n_2 (Monte Carlo, 10^6 draws). Sensitivity, specificity, NPV and accuracy are those of the reconstructed table.

Quantity	Mean	SD	2.5%	Median	97.5%
Sensitivity	0.971	0.072	0.769	1.000	1.000
Specificity	0.609	0.173	0.254	0.620	0.903
NPV	0.984	0.051	0.866	1.000	1.000
Accuracy	0.690	0.114	0.481	0.688	0.907
Prevalence	0.195	0.073	0.058	0.195	0.329
$N = n_1 + n_2$	611.3	461.3	296	492	1640

The posterior mean specificity is 0.61, with 95% credible interval $(0.25, 0.90)$. Dividing the posterior mean of TN by that of n_2 would give $364/514 \approx 0.71$, which overstates the specificity because the posterior of n_2 is strongly right-skewed. The sensitivity and NPV are high but rest mainly on the priors, which place much of the posterior mass on few or no false negatives.

The PPV, $93/243 \approx 0.38$, is determined by the observed row and does not depend on the reconstruction.

Remark on identifiability. The Svirsky case is the more weakly identified of the two examples because N is entirely unknown. The posterior for the non-diseased denominator in particular is driven largely by the prior, and summaries of n_2 and of the derived specificity, NPV and accuracy should be treated as prior-informed summaries rather than data-driven estimates. This is a setting which illustrates the importance of transparent prior specification and sensitivity analysis.

S2. Case 2: Incomplete Reporting with Known N

Data

Wismüller and Stockmaster [2] evaluated an AI-based system for detecting intracranial hemorrhage (ICH) on emergent head CT scans. The publication reports TP = 105 and FP = 17

among the 122 AI-positive cases and a total of $N = 620$ scans. The negative-class counts are not reported (Table S6). The study also reports overall sensitivity, specificity and accuracy of 95.0%, 96.7% and 96.4%, which we use in the second analysis below.

Table S6: Partially observed 2×2 table from the counts reported by Wismüller and Stockmaster [2].

	ICH Present	ICH Absent
AI positive (flagged)	105 (TP)	17 (FP)
AI negative (not flagged)	not reported (FN)	not reported (TN)

The PPV can be calculated directly as $105/122 \approx 0.86$. Because $N = 620$ is known, the counts restrict the diseased total to the compatible range $105 \leq n_1 \leq 603$, that is $0 \leq \text{FN} \leq 498$. The sensitivity lies in $[0.174, 1]$ and specificity in $[0, 0.967]$.

Analysis with the Reported Counts Only

We apply the joint TP/FP model with fixed N (Model 3, Section 4.2 of the main manuscript):

$$\text{TP} \mid n_1, p_1 \sim \text{Bin}(n_1, p_1), \tag{S6}$$

$$\text{FP} \mid n_2, p_2 \sim \text{Bin}(n_2, p_2), \tag{S7}$$

$$n_1 \mid p_3, r \sim \text{NegBin}(p_3, r)\, \mathbf{1}\{\text{TP} \leq n_1 \leq N - 1\}, \tag{S8}$$

$$n_2 = N - n_1, \tag{S9}$$

$$p_1 \sim \text{Beta}(a_1, b_1), \quad p_2 \sim \text{Beta}(a_2, b_2), \quad p_3 \sim \text{Beta}(a_3, b_3), \tag{S10}$$

$$r \sim \text{Gamma}(0.1, 0.01), \tag{S11}$$

with $a_1 = 1$, $b_1 = 0.1$, $a_2 = 0.1$, $b_2 = 1$, $a_3 = 0.1$, $b_3 = 0.5$, as in Section 6.3 of the main manuscript. The prior on p_1 places substantial mass near one, and the prior on p_2 favors a low false positive rate. The posterior was computed exactly by enumeration over n_1 (Section S4). Table S7 reports the posterior summary.

Table S7: Posterior summary: Wismüller and Stockmaster joint model (Model 3) with fixed $N = 620$, counts only, computed exactly by enumeration. Summaries of p_3 and r were computed on the refined grid described in Section S4.

Parameter	Mean	SD	2.5%	Median	97.5%
n_1	105.70	11.55	105.0	105.0	107.0
n_2	514.30	11.55	513.0	515.0	515.0
p_1	0.996	0.032	0.975	1.000	1.000
p_2	0.033	0.010	0.020	0.033	0.050
p_3	0.669	0.303	0.016	0.760	0.999
r	6.40	22.47	$<10^{-3}$	0.030	64

The posterior places probability 0.94 on $n_1 = 105$, that is on FN = 0 and a sensitivity of one, with a 95% credible interval for FN of $(0, 2)$. As in the breast MRI example shown in the main manuscript, this concentration is produced by the priors, which favor high sensitivity, rather than by the counts, which allow any value of FN from 0 to 498. Models 1 and 2 with the same priors on (p_1, p_2) give posterior means for FN of 44.8 and 24.1, with 95% intervals $(0, 392)$ and $(0, 248)$ (Table 16 of the main manuscript). With the counts alone, the reconstruction of this table is therefore determined mainly by the choice of prior.

Analysis with the Reported Sensitivity, Specificity and Accuracy

Table S8 lists the tables with TP = 105, FP = 17 and $N = 620$ whose sensitivity is closest to the reported 95.0%. No table reproduces all three reported summaries after rounding to one decimal place. The two closest, with FN = 5 and FN = 6, bracket the reported sensitivity and accuracy, and both reproduce the reported specificity. A search over all tables with $N = 620$, regardless of TP and FP, likewise found none matching all three summaries, which suggests that the summaries were computed on a slightly different set of cases than the counts. We therefore used the soft compatibility weights of Section 4.5 of the main manuscript, with $\tau = 0.005$.

Table S8: Tables compatible with the Wismüller and Stockmaster counts near the reported summaries.

n_1	FN	TN	Sensitivity	Specificity	Accuracy
110	5	493	95.5%	96.7%	96.5%
111	6	492	94.6%	96.7%	96.3%
Reported			95.0%	96.7%	96.4%

With the reported summaries, the posterior mean of FN is 5.5 with 95% interval $(4, 7)$ under Models 1 and 2, and 5.4 with interval $(4, 6)$ under Model 3. Unlike the counts-only analysis, the result is essentially the same under all three priors. Table S9 gives the reconstructed table.

Table S9: Reconstructed 2×2 table for Wismüller and Stockmaster [2] using the reported counts and summaries. The negative row shows the two most probable tables, which together carry posterior probability 0.94 under Model 1; they are not recovered counts, and the 95% credible interval for FN is $(4, 7)$ under Models 1 and 2 and $(4, 6)$ under Model 3.

	ICH Present	ICH Absent	Total
AI positive	105 (TP, observed)	17 (FP, observed)	122
AI negative	5–6 (FN, estimated)	492–493 (TN, estimated)	498
Total	$n_1 \approx$ 110–111	$n_2 \approx$ 509–510	620

Over the 95% credible interval for FN under Models 1 and 2, the sensitivity ranges from 0.94 to 0.96 and the specificity is 0.967 to three decimals. The PPV, $105/122 \approx 0.86$, is

observed, and the NPV is approximately 0.99. Approximately 5 or 6 hemorrhages were missed by the AI system among the 620 scans, conditional on the assumed compatibility tolerance (Table 17 of the main manuscript).

Comparison with Case 1

The two examples differ in how much information the report provides. In Case 1 (Svirsky), with no bound on N, the non-diseased denominator is essentially determined by the prior, and its credible interval spans nearly an order of magnitude. In Case 2 (Wismüller and Stockmaster), the known $N = 620$ bounds the missing cells, but the counts alone still allow any number of false negatives from 0 to 498, and the three priors of the main manuscript lead to very different reconstructions. It is the additional summaries reported by Wismüller and Stockmaster that help reduce the reconstruction to one or two candidate tables. This illustrates the general recommendation of the main manuscript: every summary available in a published report should be used, since a single rounded rate can be far more informative than the choice of prior.

The clinical implications also differ. For Svirsky, the wide uncertainty in specificity and NPV reflects genuine ignorance about what happened to patients with normal brush results, and no amount of modeling can resolve that without additional data. For Wismüller and Stockmaster, the reported summaries allow a precise estimate of the number of missed ICH cases, a safety-critical quantity not available directly from the reported counts.

S3. Simulation Study: Design and Detailed Results

Design

The 36 scenarios were formed by $N \in \{100, 500, 1000\}$, prevalence $\in \{0.1, 0.3, 0.5\}$, sensitivity $\in \{0.7, 0.9\}$ and specificity $\in \{0.7, 0.9\}$. In each scenario the diseased total was fixed at $n_1 = \text{round}(N \times \text{prevalence})$ and $n_2 = N - n_1$. Each replicate table was generated as

$$\text{TP} \sim \text{Bin}(n_1, \text{Se}), \qquad \text{FP} \sim \text{Bin}(n_2, 1 - \text{Sp}),$$

with $\text{FN} = n_1 - \text{TP}$ and $\text{TN} = n_2 - \text{FP}$. Replicates with $\text{TP} = 0$ would have been discarded, but none occurred, so every scenario contributed all 500 tables.

Every table was analyzed from TP, FP and N under Models 1–3 of Section 4.2 with two specifications for the accuracy priors: "uniform," $p_1, p_2 \sim \text{Beta}(1, 1)$, and "directional," $p_1 \sim \text{Beta}(1, 0.1)$ and $p_2 \sim \text{Beta}(0.1, 1)$, the priors of Section 6.3. The denominator priors were those used throughout the main manuscript: for Model 2, $\lambda \sim \text{Gamma}(1, 2/N)$ (rate parameterization, mean $N/2$); for Model 3, $p_3 \sim \text{Beta}(0.1, 0.5)$ and $r \sim \text{Gamma}(0.1, 0.01)$. Posteriors were computed by exact enumeration (Section S4), with a grid of 60×60 points for (p_3, r) and 400 points for λ.

In the reported-sensitivity experiment, the sensitivity of the true table, TP/n_1, was rounded to the nearest whole percent and imposed as a hard compatibility constraint with half-width $h = 0.005$ (Section 4.5 of the main manuscript).

Coverage is that of the 95% equal-tailed credible interval for FN. Because every derived measure is a monotone function of n_1 given TP, FP and N, the coverage is the same for all derived measures. Root mean squared error (RMSE) was computed within each scenario and then averaged over scenarios. The design of the separate reporting-discrepancy experiment is given in full in Section 5 of the main manuscript (random seed 271828). Per-scenario results for all quantities are produced by the code in the repository listed in the Data Availability Statement.

Results

Table S10: Coverage of 95% credible intervals for FN by sample size, counts only.

Model	Accuracy priors	$N = 100$	$N = 500$	$N = 1000$
1	uniform	0.963	0.959	0.958
2	uniform	0.994	0.996	0.997
3	uniform	1.000	1.000	1.000
1	directional	1.000	1.000	1.000
2	directional	1.000	1.000	1.000
3	directional	0.482	0.017	0.001

Table S11: Coverage of 95% credible intervals for FN by prevalence and by true sensitivity and specificity, counts only.

		Prevalence			(Se, Sp)			
Model	Accuracy priors	0.1	0.3	0.5	(0.7, 0.7)	(0.7, 0.9)	(0.9, 0.7)	(0.9, 0.9)
1	uniform	0.89	0.99	1.00	1.00	1.00	0.97	0.87
2	uniform	0.99	1.00	1.00	1.00	1.00	1.00	0.99
3	uniform	1.00	1.00	1.00	1.00	1.00	1.00	1.00
1	directional	1.00	1.00	1.00	1.00	1.00	1.00	1.00
2	directional	1.00	1.00	1.00	1.00	1.00	1.00	1.00
3	directional	0.29	0.13	0.08	0.07	0.08	0.24	0.27

Under the directional priors, Model 3 attains its highest coverage when the true sensitivity is 0.9, where the true number of false negatives is closest to zero, the value favored by these priors.

Table S12: Root mean squared error of the posterior mean of derived measures, computed within each scenario and averaged over the 36 scenarios, counts only.

Model	Accuracy priors	Sensitivity	Specificity	NPV
1	uniform	0.337	0.201	0.425
2	uniform	0.273	0.149	0.323
3	uniform	0.122	0.091	0.160
1	directional	0.118	0.017	0.072
2	directional	0.135	0.013	0.071
3	directional	0.202	0.017	0.104

Small RMSE values in Table S12 should be read together with coverage. Under the directional priors, for example, the RMSE for specificity is small even for Model 3, whose intervals rarely contain the truth. When the non-diseased stratum is large, an error of a few false negatives changes the estimated specificity only slightly, whereas it changes the estimated sensitivity considerably.

When the rounded sensitivity is also reported, all six specifications give the same results: coverage one, average interval width for FN of 0, 1.07 and 2.63 at $N = 100$, 500 and 1000, and FN determined exactly in 57% of replicates overall.

S4. Computation

Exact Enumeration with Known N

For the known-N models, n_1 ranges over the compatible range $\mathrm{TP} \le n_1 \le N - \mathrm{FP}$. Integrating p_1 and p_2 out of the binomial likelihood gives beta-binomial terms, so that

$$p(n_1 \mid \mathrm{TP}, \mathrm{FP}, N) \propto \pi(n_1)\,\mathrm{BB}(\mathrm{TP} \mid n_1, a_1, b_1)\,\mathrm{BB}(\mathrm{FP} \mid N - n_1, a_2, b_2). \tag{S12}$$

For Model 1, $\pi(n_1)$ is constant. For Model 2,

$$\pi(n_1) = \int \frac{\mathrm{Poisson}(n_1 \mid \lambda)}{P_\lambda(1 \le X \le N-1)}\,\mathrm{Gamma}(\lambda \mid a_\lambda, b_\lambda)\,d\lambda, \tag{S13}$$

and for Model 3,

$$\pi(n_1) = \iint \frac{\mathrm{NegBin}(n_1 \mid p_3, r)}{P_{p_3,r}(\mathrm{TP} \le X \le N-1)}\,\mathrm{Beta}(p_3 \mid a_3, b_3)\,\mathrm{Gamma}(r \mid a_r, b_r)\,dp_3\,dr. \tag{S14}$$

The truncations are normalized, so that each prior is a proper distribution on its support, as the model specification states. The integrals were evaluated on equal-probability grids of the prior distributions (400 points for λ; 60×60 points for (p_3, r) in the simulation and 200×200 points for the empirical tables), with all table quantities (n_1, n_2, FN, TN and the derived measures) and the summaries of p_1 and p_2 unchanged to the reported precision when the grids were refined. The posterior summaries of p_3 and r themselves are not stable on such coarse grids, because an equal-probability grid represents the extreme right tail of

the Gamma$(0.1, 0.01)$ prior on r by very few points. They were therefore computed on a refined grid of 1000×1500 points, equal-probability in p_3 and equally spaced in $\log r$ over $(10^{-200}, 3000)$, with quantiles obtained by linear interpolation of the cumulative posterior within grid cells; doubling both dimensions left them unchanged to the reported precision. Posterior summaries of p_1 and p_2 follow from the beta posteriors given n_1. Reported summaries enter by multiplying the posterior of each table by the indicator or soft compatibility weight of Section 4.5 of the main manuscript. All calculations were carried out in R.

Single-row model. For the single-row model, p is integrated out analytically, giving $p(n \mid y) \propto \mathrm{BB}(y \mid n, \alpha, \beta)\, \pi(n)$, where

$$\pi(n) = \sum_{r \geq 1} P(r) \int \frac{\mathrm{NegBin}(n \mid p^\star, r)}{P_{p^\star, r}(X \geq y)}\, \mathrm{Beta}(p^\star \mid \alpha^\star, \beta^\star)\, dp^\star, \tag{S15}$$

with $P(r)$ the Poisson–gamma (negative-binomial) marginal of r restricted to $r \geq 1$, and the normalizing probability taken over $y \leq X \leq N$ when an upper bound N is imposed. The integral over $p^\star$ was evaluated on an equal-probability grid of 300 points, and the sum over n was truncated at a bound (5,000 for the diseased strata and 40,000 to 60,000 for the non-diseased strata of the breast MRI and Svirsky examples) beyond which the posterior mass was below 10^{-6}. Posterior summaries of p and λ follow from their conditional distributions, $p \mid n \sim \mathrm{Beta}(y + \alpha, n - y + \beta)$ and $\lambda \mid r \sim \mathrm{Gamma}(a + r, b + 1)$.

Extension to Unknown N

When N is not reported, the table space is no longer finite, and an upper bound $N_{\max}$ and a prior on N are required. The posterior over the joint states (N, n_1) is

$$p(N, n_1 \mid \mathrm{TP}, \mathrm{FP}) \propto \pi(N)\, \pi(n_1 \mid N)\, \mathrm{BB}(\mathrm{TP} \mid n_1, a_1, b_1)\, \mathrm{BB}(\mathrm{FP} \mid N - n_1, a_2, b_2), \tag{S16}$$

over $\mathrm{TP} + \mathrm{FP} \leq N \leq N_{\max}$ and $\mathrm{TP} \leq n_1 \leq N - \mathrm{FP}$, where $\pi(n_1 \mid N)$ is the Model 1, 2 or 3 prior for that value of N. This is a finite double sum, so the same enumeration applies, with at most of the order of $N_{\max}^2/2$ candidate tables. Reported summaries enter exactly as in the known-N case.

The general function `complete_2x2` in the code repository implements this extension for a Dirichlet–multinomial prior on the table: given N, the cell counts have a Dirichlet–multinomial distribution with parameter $\boldsymbol{\alpha}$ (uniform, $\boldsymbol{\alpha} = (1, 1, 1, 1)$, by default), and N has either a log-uniform prior, $\pi(N) \propto 1/N$ (the default), or a uniform prior on $\{1, \ldots, N_{\max}\}$. The function enumerates every table consistent with the reported cells and margins with total at most $N_{\max}$. Because the posterior can depend strongly on both $N_{\max}$ and $\pi(N)$, these choices should always be reported and varied. The single-row model of Section 4.1 is the alternative used in the main manuscript and in Section S1: it requires no upper bound because the negative-binomial prior on each stratum total is proper.

WinBUGS Implementation

WinBUGS code for Models 1–3 is available in the code repository; all results reported in the main manuscript and in this supplement for Models 1–3 were computed by exact enu-

meration. We also implemented a separate model in WinBUGs for an independent check of the enumeration with reported summaries in which the table has a Dirichlet prior on the four cell probabilities and a multinomial likelihood. The multinomial was written as a sequence of binomials, so that TP and FP are observed and FN is a latent node, with $\mathrm{TN} = N - \mathrm{TP} - \mathrm{FP} - \mathrm{FN}$. Each reported summary enters through the "ones trick": an auxiliary Bernoulli observation equal to one whose success probability is the rounding indicator of Section 4.5 of the main manuscript, or, when no table reproduces all reported summaries, the soft compatibility weight with $\tau = 0.005$. The same Dirichlet–multinomial model was also computed exactly by enumeration, which provides the reference values for the MCMC results. The WinBUGS model file and the R scripts used to run it are available in the code repository.

WinBUGS Results

Table S13 compares the WinBUGS posterior summaries of FN with the exact values for the same model (uniform Dirichlet prior). WinBUGS version 1.4.3 was run from R through the R2WinBUGS package with three chains, each of 55,000 iterations including 5,000 burn-in, thinned by a factor of 150, giving 1,002 retained draws in total. The first example is a constructed table with TP = 48, FP = 12 and $N = 1000$ and a reported accuracy of 97% to the nearest whole percent, which 11 of the 941 compatible tables reproduce.

Table S13: Posterior summaries of FN for the Dirichlet–multinomial model: WinBUGS and exact enumeration.

	Exact			WinBUGS		
Example	Mean	Median	95% interval	Mean	Median	95% interval
Constructed[a], accuracy 97%	18.00	18	(13, 23)	18.14	18	(13, 23)
Breast MRI, Se 96%	3.00	3	(3, 3)	3.00	3	(3, 3)
ICH study, Se/Sp/Acc	5.53	6	(5, 7)	5.56	6	(5, 7)
ICH study, counts only	249.0	249	(12, 486)	262.2	271.5	(12, 491)

[a] TP = 48, FP = 12, $N = 1000$. ICH study: Wismüller and Stockmaster [2].
$\hat{R} \leq 1.003$ for FN in all four WinBUGS runs.

In the first three examples, the WinBUGS medians and 95% intervals coincide with the exact values. In the counts-only Wismüller and Stockmaster analysis, where the posterior is spread over almost 500 values, the WinBUGS mean of 262.2 differs from the exact value of 249.0 by about 2.5 Monte Carlo standard errors (the MC error reported by WinBUGS is 5.2), which reflects the small number of retained draws relative to the spread of the posterior. The Dirichlet–multinomial results with reported summaries are close to those of Models 1–3 in Table 16 of the main manuscript (posterior mean 5.5 with interval $(4, 7)$ under Models 1 and 2), whereas the counts-only results differ substantially from those of Models 1–3 because, without the summaries, the reconstruction is determined mainly by the prior.

S5. Full Posterior Summaries for the Breast MRI Application

This section gives the full posterior summaries, including the hyperparameters, for the breast MRI benchmark of Section 6 of the main manuscript (Langlotz [3]; TP = 71, FP = 28, FN = 3, TN = 80, $N = 182$), together with the results under priors chosen by the rule of Section 4.4. All values were computed exactly as described in Section S4.

Single-Row Model

Table S14 extends Tables 7, 8, 10 and 11 of the main manuscript to all model parameters. The diseased-stratum analyses use $a = 1$, $b = 0.1$, $\alpha = 2$, $\beta = 1$, $\alpha^\star = 1$, $\beta^\star = 1$, and the non-diseased-stratum analyses use $a = 2$, $b = 1$, $\alpha = 2$, $\beta = 5$, $\alpha^\star = 1$, $\beta^\star = 50$. The "upper bound" analyses impose $n \le N = 182$ within each stratum separately, without the joint identity $n_1 + n_2 = N$ (Section 6.2).

Table S14: Posterior summaries for all parameters of the single-row model, breast MRI example (true $n_1 = 74$, $n_2 = 108$).

Analysis	Parameter	Mean	SD	2.5%	Median	97.5%
Diseased, no bound	λ	10.25	9.41	0.75	7.43	35.48
	n_1	74.58	16.66	71	71	98
	p	0.954	0.081	0.717	0.980	0.999
	$p^\star$	0.555	0.265	0.082	0.562	0.978
	r	10.27	9.80	1	7	37
Non-diseased, no bound	λ	2.182	1.308	0.433	1.920	5.399
	n_2	134.2	138.2	41	98	443
	p	0.301	0.154	0.063	0.281	0.645
	$p^\star$	0.023	0.017	0.003	0.019	0.069
	r	2.366	1.581	1	2	6
Diseased, $n_1 \le N$	λ	10.31	9.47	0.76	7.47	35.72
	n_1	74.98	12.64	71	71	109
	p	0.950	0.090	0.646	0.979	0.999
	$p^\star$	0.549	0.269	0.065	0.558	0.978
	r	10.34	9.87	1	7	37
Non-diseased, $n_2 \le N$	λ	2.381	1.442	0.468	2.089	5.960
	n_2	107.2	40.22	43	104	178
	p	0.303	0.131	0.138	0.270	0.624
	$p^\star$	0.017	0.017	0.001	0.013	0.062
	r	2.762	1.877	1	2	8

Joint Model with Fixed N

Table S15 extends Table 12 of the main manuscript to the hyperparameters of the Model 3 denominator prior. The priors are $p_1 \sim \text{Beta}(1, 0.1)$, $p_2 \sim \text{Beta}(0.1, 1)$, $p_3 \sim \text{Beta}(0.1, 0.5)$ and $r \sim \text{Gamma}(0.1, 0.01)$, and (p_3, r) were integrated as described in Section S4. The posterior probability of $n_1 = 71$, that is FN $= 0$, is 0.93.

Table S15: Posterior summary for all parameters of the joint fixed-N Model 3, breast MRI example (true $n_1 = 74$). Summaries of p_3 and r were computed on the refined grid described in Section S4.

Parameter	Mean	SD	2.5%	Median	97.5%
n_1	71.48	3.91	71	71	74
n_2	110.52	3.91	108	111	111
p_1 (sensitivity)	0.994	0.034	0.952	1.000	1.000
p_2 (false positive rate)	0.252	0.046	0.176	0.250	0.339
p_3	0.649	0.321	$<10^{-3}$	0.747	0.999
r	5.69	20.86	$<10^{-3}$	0.024	57

The posteriors for p_3 and r are very dispersed, and, as discussed in Section 6.3 of the main manuscript, the concentration of n_1 at its lower bound comes mainly from the $\text{Beta}(1, 0.1)$ prior on p_1.

Priors Chosen by Rule

The rule of Section 4.4 centers the accuracy priors on the meta-analytic estimates $\text{Se}_0 = 0.90$ and $\text{Sp}_0 = 0.72$ of Peters et al. [4], giving

$$p_1 \sim \text{Beta}(0.90m,\, 0.10m), \qquad p_2 \sim \text{Beta}(0.28m,\, 0.72m)$$

for prior concentration m. In the single-row model both strata use the Step 3 denominator priors, $p^\star \sim \text{Beta}(1, 1)$ and $\lambda \sim \text{Gamma}(1, 0.1)$. Table S16 extends Table 13 of the main manuscript with standard deviations, medians, and the diseased stratum under the upper bound $n_1 \le N$.

Table S16: Posterior summaries of the stratum totals in the single-row model with priors chosen by rule, breast MRI example (true $n_1 = 74$, $n_2 = 108$). "Bound" denotes the single-stratum upper bound $n \le N = 182$.

Prior on p	Quantity	Mean	SD	2.5%	Median	97.5%
Rule, $m = 5$	n_1	71.6	2.4	71	71	76
	n_1, bound	71.6	2.6	71	71	76
	n_2	93.6	126.2	29	63	336
	n_2, bound	85.8	48.0	29	70	180
Rule, $m = 10$	n_1	72.2	2.7	71	71	79
	n_1, bound	72.2	2.9	71	71	79
	n_2	99.4	69.6	38	81	270
	n_2, bound	101.5	44.2	39	92	180
Rule, $m = 20$	n_1	73.1	3.2	71	72	82
	n_1, bound	73.1	3.3	71	72	82
	n_2	99.5	44.2	48	89	212
	n_2, bound	107.5	39.5	49	100	180
Uniform, Beta$(1, 1)$	n_1	78.2	48.8	71	72	125
	n_2	34.3	41.7	28	29	76

Table S17 gives the corresponding results for the joint fixed-N analysis under Models 1–3, with the denominator priors of Section 6.3 for Model 3 ($p_3 \sim \text{Beta}(0.1, 0.5)$, $r \sim \text{Gamma}(0.1, 0.01)$, integrated on a 100×100 grid) and $\lambda \sim \text{Gamma}(1, 2/N)$ for Model 2. All nine 95% credible intervals contain the true value FN = 3, and the posterior medians lie between 0 and 7. Results for Model 1 over a wider range of m are given in Table 14 of the main manuscript.

Table S17: Posterior summaries of FN in the joint fixed-N analysis with priors chosen by rule, breast MRI example (true FN = 3).

m	Model	Mean	Median	95% interval
5	1	10.5	5	(0, 51)
	2	8.7	4	(0, 44)
	3	1.8	0	(0, 18)
10	1	9.7	6	(0, 38)
	2	8.6	6	(0, 34)
	3	2.9	1	(0, 20)
20	1	8.9	7	(0, 28)
	2	8.4	7	(0, 26)
	3	4.3	2	(0, 19)

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